

CARNOT RESEARCH Pvt Ltd, IIT Delhi

IILM UNIVERSITY, GN



INTERNSHIP PRESENTATION

ON-PREMISE RAG AND AGENTIC SOLUTIONS

Presented by MUDIT AGRAWAL

B.Tech CSE(AI/ML) | 2026-27 | 4CSE1 | CS-2341718

INTERNSHIP AT GLANCE

ORGANISATION

Carnot Research Pvt Ltd -
An IITD incubated startup



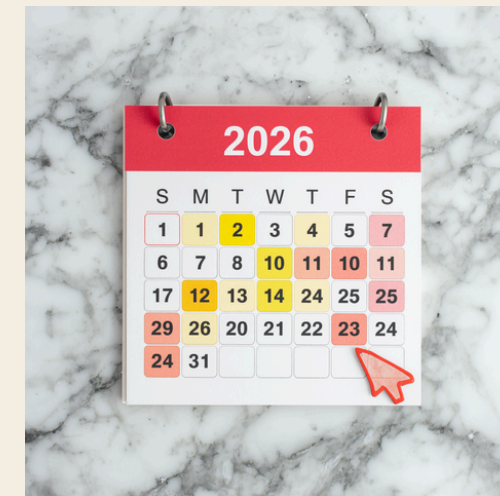
ROLE

AI & ML Intern



Duration

June 01 2026 to
August 31 2026



Focus

End-to-End On-
Premise RAG and
Agentic AI
workflows



ORGANISATION PROFILE

Carnot Research Pvt Ltd

Founded by IIT Delhi researchers, Carnot Research Pvt. Ltd. is a deep-tech company specializing in production-grade Generative AI, Computer Vision, and NLP. Carnot builds 100% on-premise, cloud-independent AI systems that ensure full data sovereignty for mission-critical environments. These research-driven solutions operate securely in air-gapped infrastructures without external data exposure. Serving defense, government, and critical enterprise sectors, Carnot is CMMI Level 3 and ISO 27001 certified, and trusted by the NSG, IIT Delhi, and BCG.



MY WORK AT CARNOT



WORK SUMMARY – INTERNSHIP JOURNEY

Carnot Research Pvt Ltd (An IITD Incubated Startup)



1st June 2026 – 26th August 2026
(87+ Days of Continuous Work)



Worked on cutting-edge AI systems, LLM agents, RAG workflows, ETL pipelines and full-stack implementations to build intelligent, production-ready solutions.



OVERALL IMPACT

- Built end-to-end AI systems from research to production
- Delivered multiple POCS & production ready pipelines
- Optimized for performance, accuracy and real-world usability
- Collaborated with senior leadership & contributed to key product decisions



87+
Days of
Continuous Work



50+
Major Tasks
Completed



10+
End-to-End
Implementations



High
Performance
& Accuracy



Cross-functional
Collaboration &
Leadership

JOURNEY TIMELINE

1 FOUNDATION & RESEARCH (1st – 12th June)



- Researched LLM datasets, benchmarks & agriculture datasets (NDAP)
- Revised CNN, LSTM, RNN architectures
- Benchmarked Gemma 4 E4B (0.51 ANLS on InfoVQA)
- Explored NLP-to-SQL agent & built evaluation framework (96% test accuracy)

Focus: Research, Benchmarking, Exploration & Evaluation

2 POC DEVELOPMENT & AGENT BUILDING (8th – 16th June)



- Built first POC using Gemini API with RAG (ChromeDB, Elasticsearch)
- Implemented document parsing, OCR (Mistral OCR), Streamlit UI
- Fixed context, errors & edge cases
- My POC selected for prototype
- Tested NDAP portal & worked on map, STT, TTS, end-to-end pipeline

Focus: RAG Agent, Ingestion, OCR, UI & Testing

3 ETL PIPELINE & DATA ENGINEERING (19th – 25th June)



- Designed standard ETL pipeline (Pandas, Spark, Airflow)
- Implemented Airflow DAG + Spark cleaning pipeline
- Made pipeline dataset-agnostic with pattern caching
- Built for scalability, reusability & automation

Focus: ETL Architecture, Data Cleaning, Automation

4 AGENTIC OS DEVELOPMENT (26th June – 3rd July)



- Built Agentic OS (Streamlit → CLI) with tools: Read, Write, Doc Gen, Remote Exec, Python Exec, Draft Emails & more
- Integrated memory, embeddings (BGM), Qwen 2.5 8B LLM
- Added NLU, improved skills, reduced compute time by 90% (lazy ingestion)

Focus: Agentic OS, Skills, CLI, Email Automation

5 EMAIL & CALENDAR AGENT (4th – 17th July)



- Built full email agent with Read, Draft, Send, Reply, attachments & time-based search
- Integrated Calendar/Meet sub-agent with OAuth, auto scheduling, slot checking & collision detection
- Human-in-the-loop confirmations for meetings

Focus: Email Automation, Calendar/Meet, Integration

6 AUTOMATION, TESTING & INTEGRATION (20th – 31st July)

- Built CI/CD & testing pipeline (RAGAS, TruLens, PyTest)
- Integrated entire email pipeline (9000+ lines) into company repo
- Automated meeting reply system in 21 phases with thread tracking & deduplication
- Stress tests, bug fixes & documentation

Focus: Automation, Integration, Testing, Documentation

7 INTEL OPTIMIZED RAG WORKFLOWS (1st – 12th August)

- Designed & implemented 3 RAG workflows (toggle-based)
- Switched to Quadrant (Intel-optimized vector DB)
- Added Docling OCR, NLU, multi-turn query handling
- Evaluated using BEIR, RAGAS, HotpotQA with Gemini 2.5 Flash

Focus: Evaluation, Optimization, RAG Workflows

8 BENCHMARKING & FINALIZATION (15th – 26th August)

- Fixed retrieval bugs, re-indexed data
- Completed BEIR, RAGAS & HotpotQA benchmarks
- Achieved 90+ retrieval & response accuracy with < 3s latency
- Finalized Advanced RAG for on-premise deployment & tested on Ollama (Qwen 2.5:8B)

Focus: Benchmarking, Optimization, On-Prem Deployment



KEY ACHIEVEMENTS



0.51
ANLS Score on
InfoVQA Benchmark
(Gemma 4 E4B)



96%
Accuracy Achieved
in Framework
Evaluation



90%
Reduction in Temporary
Workspace Compute Time
(Lazy Ingestion)



< 3s
Query Latency Achieved
in Optimized RAG
Workflows



90+
Retrieval & LLM Response
Accuracy on Benchmarks
(BEIR, RAGAS, HotpotQA)



9000+
Lines of Code Integrated
into Company Repository
(Phased Integration)

TECH STACK & TOOLS

LLMs & APIs: Gemini 2.5 Flash, Qwen 2.5 (8B/7B), Gemma 4 E4B, Mistral OCR, Groq Cloud
Frameworks & Libraries: LangChain, LlamaIndex, Streamlit, Pandas, Spark, Airflow, Docling, ReportLab, WeasyPrint, HTMLDOCX
Databases & Vector Stores: PostgreSQL, ChromaDB, Quadrant (Intel-Optimized)
Evaluation & Tools: RAGAS, BEIR, HotpotQA, TruLens, PyTest, OpenVINO IR, Docker, Ngrok, OAuth 2.0

WHAT I BUILT

- 🔍 **RAG Pipelines (3 Workflows)** – Advanced, Modular & Optimized for Intel Hardware
- ✉️ **Intelligent Email Agent** – Read, Draft, Send, Reply, Categorize, Attachments, Time Search
- 📅 **Calendar/Meet Agent** – Auto Scheduling, Collision Detection, Meet Link Generation
- 🔄 **ETL Pipeline** – Dataset-Agnostic Cleaning Pipeline with Pattern Caching
- 🤖 **Agentic OS** – Multi-Skill System with CLI/Streamlit Interface
- 📊 **Evaluation Framework** – Benchmarking with BEIR, RAGAS, HotpotQA
- 📄 **Document Processing** – OCR, Infographic Extraction, PDF/DOC Generation



KEY TAKEAWAYS

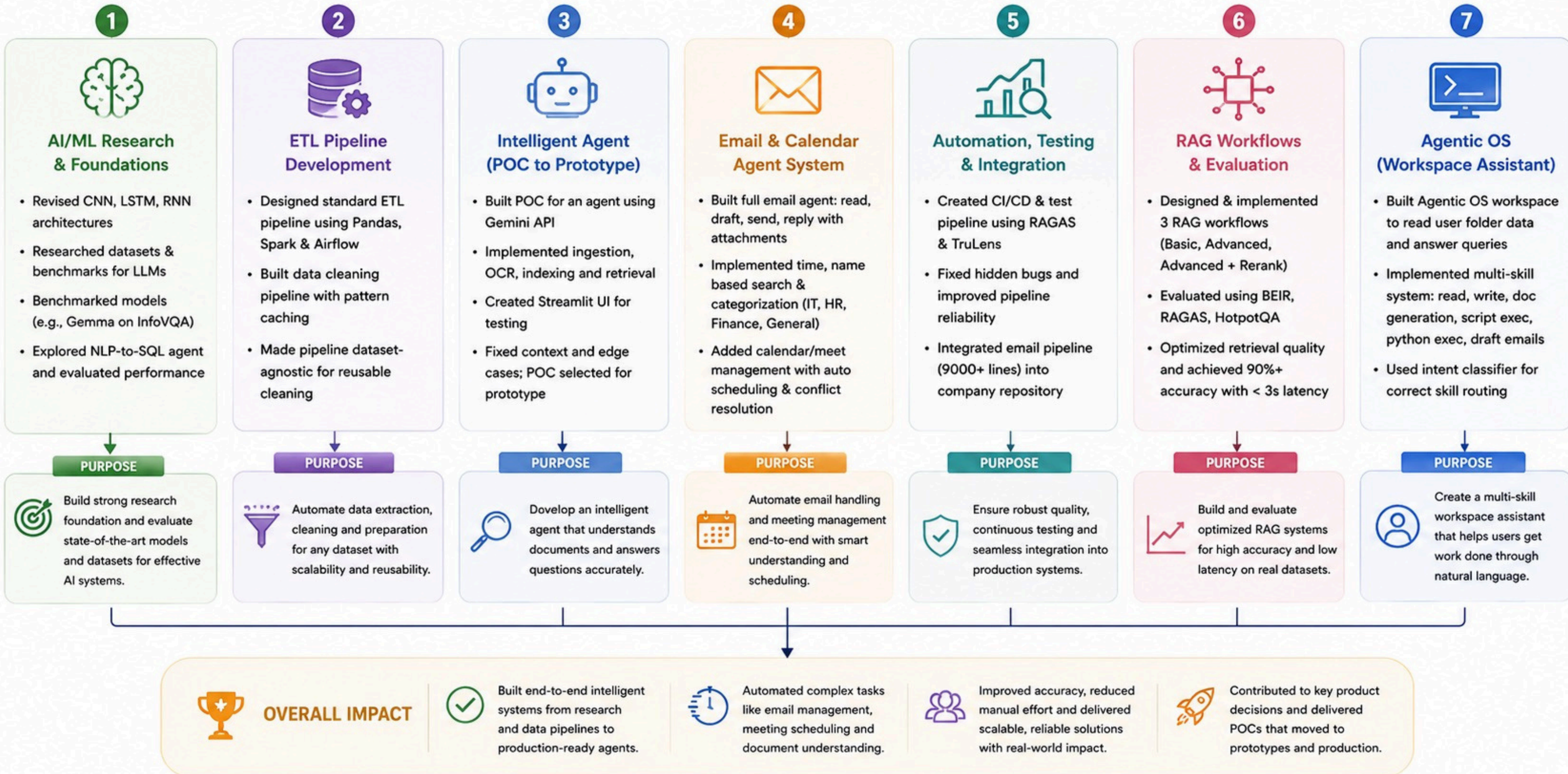
- ✓ End-to-end ownership of complex AI systems from POC to production.
- ✓ Strong focus on accuracy, latency & user experience.
- ✓ Built scalable, modular & maintainable architectures.
- ✓ Collaborated with leadership & drove critical decisions.
- ✓ Delivered production-ready solutions with real impact.



GRATEFUL FOR THE OPPORTUNITY!

Thankful to Carnot Research Pvt Ltd and the entire team for the guidance, trust and support throughout this incredible learning journey.

PROJECTS I WORKED ON – OVERVIEW & PURPOSE



TECH STACK I USED AT CARNOT

TECH STACK & TOOLS I USED AT CARNOT RESEARCH



A collection of technologies, frameworks, databases, tools and platforms I worked with

PROGRAMMING LANGUAGES


Python


SQL


PySpark


pandas

AI / ML & DEEP LEARNING


PyTorch


TensorFlow


scikit-learn


OpenAI

LLMs & MODELS


Gemini


Gemma


Qwen


Mistral AI


BAI (BGE)


Ollama

DATA & VECTOR STORES


PostgreSQL


ChromaDB


Elasticsearch


Quadrant

RAG & EVALUATION


LangChain


LlamaIndex


Streamlit


RAGAS


TruLens


BEIR
Benchmarking Information Retrieval

FRAMEWORKS & LIBRARIES


Airflow


FastAPI


NLTK


spaCy


ReportLab


WeasyPrint


Docling


Unstructured


OCR (Various)


OAuth 2.0


ReAct Workflow

DOCUMENT & FILE PROCESSING


pdf2docx


htmldocx


python-docx


Markdown


Nu Extract

PLATFORMS & CLOUD


Google Cloud


Groq Cloud


Hugging Face


GitHub

INFRASTRUCTURE & TOOLS


Docker


ngrok


Git


CI/CD


pytest

UI & INTERFACE


Streamlit


CLI

OTHERS


Chrome


IntelliJ IDEA


VS Code

I leveraged these technologies to

Build intelligent agents and multi-skill systems

Develop robust data pipelines and automate workflows

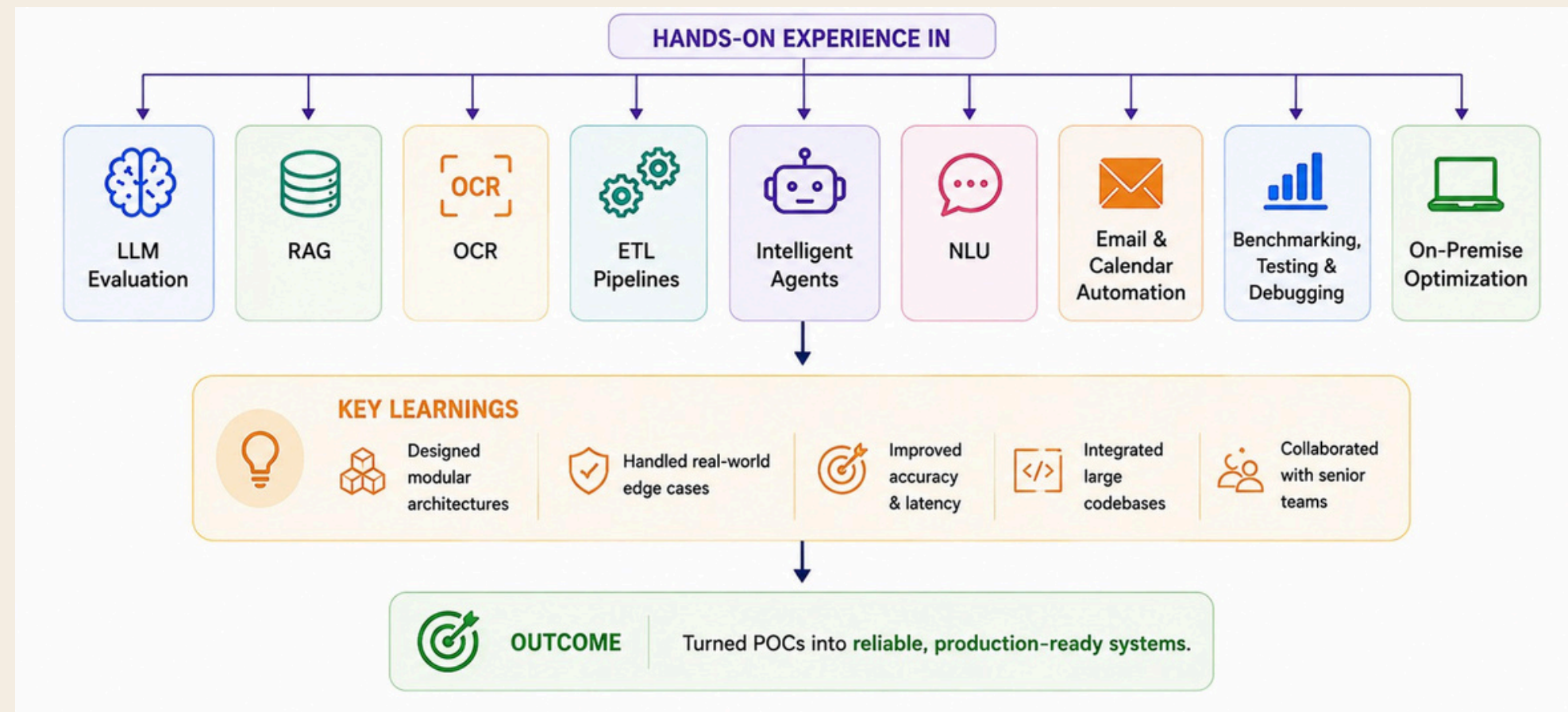
Implement RAG systems with high accuracy & low latency

Create user-friendly interfaces and production-ready solutions

Continuously test, evaluate and optimize performance

WHAT I LEARNED

During my time at Carnot Research, I learned how to take AI systems from research and experimentation to production. I gained hands-on experience in LLM evaluation, RAG, OCR, ETL pipelines, intelligent agents, NLU, email and calendar automation, benchmarking, testing, debugging, and on-premise optimization. I learned to design modular architectures, handle real-world edge cases, improve accuracy and latency, integrate large codebases, and collaborate with senior teams to turn POCs into reliable, production-ready systems.



THANK YOU